Embedded system Concept

* ① Computing system Definition
  - device → some operation (multitask)
  - Computing system Component (
    - Processor → Multi Core
    - Memory → RAM :. by giga / ROM :. by Tera
    - I/O Peripherals)

② Embedded system
- it's Computing system with limit Resource
  - Processor: single Core / Double Core
  - Memory: ROM 32kB, RAM 2kB
  - I/O limited
- to do specific task

③ Embedding system
- it's multi Embedded systems communicate together

* Compare SoC Vs SoB

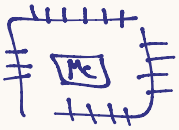
Factors	SoC	SoB
Performance	—	—
Size	Memory :. — Volume :. low	Memory :. — Volume :. High
Cost	Low	High
Power Consumption	Low	High
Configurability	N/A Production after Design	Easy For develop & Design Phase

* Mc → MicroController → SoC → Process, Memory, I/O
* ECU → Electronic Control unit → electronic Component

* Embedded Challenges

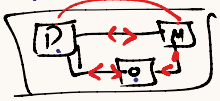
1. Performance → target High
2. Cost → low
3. Size (Memory/Volume) → low
4. Power Consumption → low

* Development kit

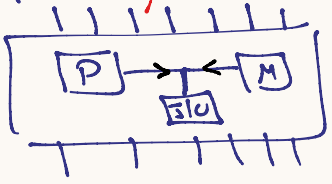


* How design Embedded system?

① system on Board (SoB)
- white Board/PCB
- Bus → wire / track
- use when H.W development

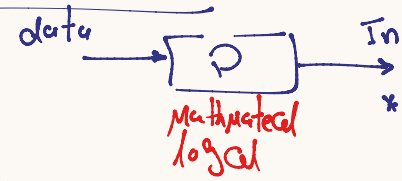


② system on chip (SoC) (Mc)
- inside chip
- Bus :. tunnel
- use when Production Phase



* Embedded system → Processor → ①
  - Memory
  - I/O Peripheral

[1] Processor



* Notes about Processor

① Processor need Power & Clock to work

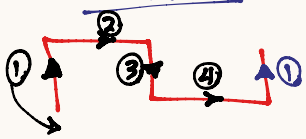
② Processor work with Memory only

③ Clock it's square wave

④ Processor → Machine cycle

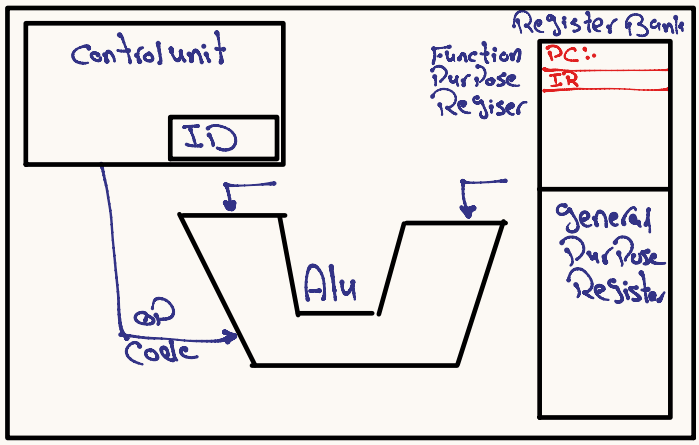
*Common

① fetch
② decode
③ execute
④ Interrupt Req



⑤ Machine cycle must be to execute Instruction

* Inside Processor



* Machine cycle sequence

[1] Fetch :. Processor get the Instruction will be executed From Memory (ROM) → (Code Memory)

- ⓐ Control unit decode the fetch time
- ⓑ Control unit Read PC Register to get the Address of Instruction.
  * PC:. Program Counter store the Instruction Address will be executed.
- ⓒ Processor will be Communicate with Address to get Instruction.
- ⓓ Control unit store the Instruction in the IR Register.
  * IR:. Instruction Register → Store Instruction
- ⓔ Control unit update the PC to next Instruction Address (1)

[2] decode :. Processor understand Instruction to get

- ① opCode [Operation]
- ② operand
- ③ who will be execute.

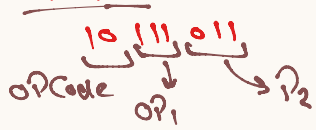
Control unit
Instruction
ID: Decoder

Forevery Processor have Instruction set

Add 011
sub 101
or 001

opCode → unique

* Instruction formate



* opCode → who will be execute

- Add → Alu
- Store → Control unit

* Processor History

① Processor → vacuum tube

② Micro Processor → transistor

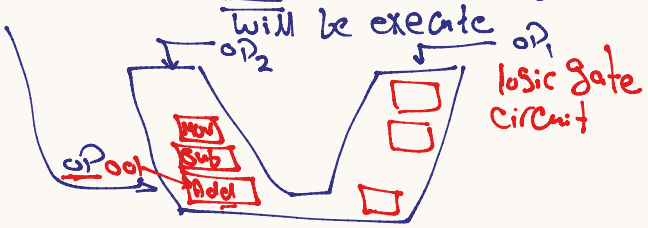
→ today :. Processor | µProcessor

③ CPU:. Centeral Processor unit

→ in Embeded system.

Single Core	Double Core
µP = CPU = Processor	CPU
	GPU

[3] execute :. Control unit will be select if his execute Instruction or the Alu → Arthimatical logical unit will be execute



ISA :. Instruction Architecture

* RISC

→ Reduce Instruction set Architecture
→ range ≃ 100 Instruction
→ Almost Instruction will be executed in 1 clk

* CISC

* Complex Instruction set Architecture.
* rang ≃ 1000 Instruction
* Almost Instruction will be executed in 4 clk cycle

① Performance (Risc)

↳ 1 clk
* 5 + 4 → 1 clk
* 10 * 100 → Not support
↳ 100 + 100 + 100 + 100 + 100
↳ 10 clk cycle

the same

① Performance (CISC)

↳ 4 clk
*ex 5 + 4 → 4 clk
*ex 10 * 100 → 4 clk

the same

② Cost (Risc)

H.w = simple → Low cost
sw = Complex → High cost

the same

② Cost (CISC)

H.w = Complex → High cost
S.w = simple → low cost

the same

③ Size (Risc)

↳ Alu → small Instruction set → small size ↓
↳ I.D :. Hardwired
↳ H.w Circuit ↑
↳ very fast
↳ 1 clk

the same

③ Size (CISC)

↳ Alu → large Instruct set → large size ↑
↳ I.D :. Programmed
↳ small chip
↳ sw Algorithm search about op Code ↓
↳ small
↳ 4 clk

the same

④ Power Consumption (Risc)

↳ work 1 circuit in Alu

the same

④ Power Consumption (CISC)

↳ work 1 Circuit in Alu

the same

ex :. Intel → Cisc ✓
ARM → Risc ✓

Apple → change Processor From Intel into ARM
↳ low Power
↳ 3.3v

Note in side Processor
FPR :. Function Purpose Register
↳ PC → Program Counter
↳ IR → Instruction Reg
↳ Acc → Accumulator
↳ PSW → Processor status word
↳ SP → Stack Pointer

[2] Memory

Volatile Memory

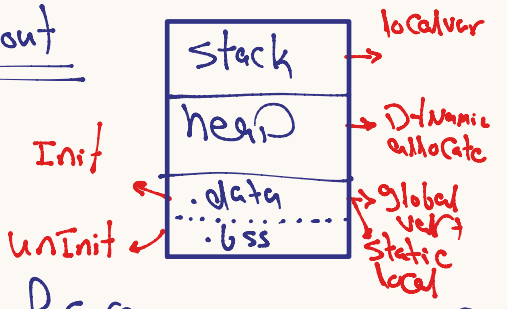
* save Power - save Data
* loss Power - loss Data
* like :. RAM (data Memory)
↳ You write data During Runtime

Non Volatile Memory

* Save Power - Save Data
* loss Power - save Data
* like :. ① ROM (Code Mem)
② NVRAM
↳ H.w NVRAM → Battery + RAM
↳ SW NVRAM → RAM

Ram :. Random Access Memory

↳ * timing Random Not location.
↳ * layout



↳ * Manfacure → DDR4, DDR5

↳ ① Based on Cap :. Dynamic RAM
↳ low cost ↳ large scale
↳ After time Cap discharge
↳ Solve → Periodic charge

↳ ② Based on transistor (Bjt)
↳ static RAM :. SRAM
↳ High Cost
↳ save value
↳ used in Embedded

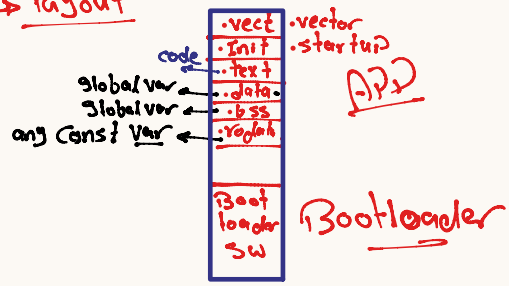
in Atmega32 → 2KB
↳ SRAM

ROM: Read only Memory.

↳ Processor can Read from this Memory

↳ Code Memory

↳ layout



↳ Manfacture :. Based on transistor (MOSFET)

↳ to write on ROM need High Power

↳ Need to Programmable kit to upload code on ROM (USBASB)

↳ ROM type

[1] Masked ROM → ×

↳ factory will be upload code.

↳ You can't change code

[2] OTP → one time Programmable (ROM)

↳ Memory is empty → from factory

↳ You can upload your code one time only.

[3] EPROM → Erasable Programmable (ROM)

↳ You can upload your code multi time

↳ You can erase ROM by (UV) → long time

[4] EEPROM → Electricty Erasable Programmable ROM

↳ You can upload your code multitime

↳ You can erase ROM by electricty.

↳ write byte by byte

↳ Can write data and Modify it During Runtime

Code

data

save data

[5] Flash ROM

↳ Erase by electricty

↳ Upload code Multi time

↳ word by word (Page | Page)

word | Page → No of byte.

↳ save code only.

Memory Conclusion

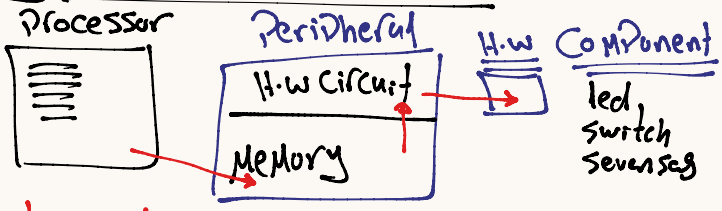
RAM:. Runtime (data)

ROM:. code

in Embeelded system.

* Code → Flash
* data → SRAM
  ↳ EEPROM

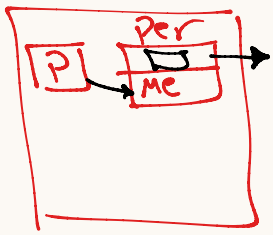
[3] IO Peripheral



* it's gateway between Processor & H.w Component

* For each H.w Component specific Peripherals
  ↳ Bluetooth → Uart → Processor
  ↳ Led → Dio → Processor

* Peripheral → inside Mc



For each Peripheral have H.w Circuit & Peripheral Memory